The Candy Factory Caper

Whodunnit? · Area, Surface Area & Volume · Teacher Guide and Answer Key

Unit	Time	Supplies	Group size
Area, Surface Area & Volume	20–40 minutes	Pencil, scrap paper, scissors	Teams of 2–3 (or solo)

Skills practiced: Area of triangles, trapezoids and circles; surface area and volume of prisms, cylinders, cones, spheres and pyramids.

General procedures

1. Give each student the **student sheet** (story and case board) and the **Show Your Work** page.
2. Print the **Clue Cards** (pages 3–7 of the student packet) and cut them apart on the dashed lines. Laminating them lets you reuse the set. Post them around the room.
3. Students visit the cards in any order. At each card they solve the problem on their work page, find their answer among the four choices, read the clue beside it, and cross things off the case board. **Every wrong choice leads to a false clue** that clears the real culprit, so a mistake shows up when the board does not narrow down to one suspect, one place and one method.
4. When a team thinks it knows, they bring you the three names. If they are wrong, let them hunt for the card where something went wrong.

Solution: Dr. Fizz took it, in the Gumdrop Garden, using this method: Hidden in a gumball machine.

The reveal (read aloud): During the factory tour, **Dr. Fizz** slipped into the **Gumdrop Garden**, hid the Golden Recipe inside a **gumball machine**, and carried the machine out as a “repair job.” Every alibi and method you cleared led to the scientist. Sweet victory!

Clue map: what each correct clue clears

Card	Clears
1	Suspect: Mr. Sprinkle · Place: Chocolate River
2	Method: Packed in a candy tin
3	Suspect: Ms. Caramel · Place: Taffy Room
4	Method: Wrapped as a lollipop
5	Suspect: Gus Barton · Place: Loading Dock
6	Method: Floated in chocolate
7	Suspect: Lolly Park · Place: Test Kitchen
8	Method: Rolled in a candy barrel
9	Suspect: Inspector Mint · Place: Packing Floor
10	Method: Slipped in a sample bag

Answer Key

The Candy Factory Caper · Correct choice and worked solution for each card

#	Problem	Correct choice	Worked solution	Clue it leads to
1	Find the area of a triangle with base 10 cm and height 6 cm.	**D.** 30 cm²	$A = \frac{1}{2}bh = \frac{1}{2}(10)(6)$ 30 cm²	Mr. Sprinkle was giving a tour by the Chocolate River. Twenty guests remember him.
2	Find the area of a trapezoid with bases 6 cm and 10 cm and height 4 cm.	**A.** 32 cm²	$A = \frac{1}{2}(b_1 + b_2)h = \frac{1}{2}(16)(4)$ 32 cm²	Every candy tin was opened at the dock. Only candy.
3	Find the area of a circle with radius 3 cm. Leave your answer in terms of π.	**B.** 9π cm²	$A = \pi r^2 = \pi(3^2)$ 9π cm²	Ms. Caramel was stretching taffy in the Taffy Room with her team.
4	Find the volume of a rectangular prism that is 3 cm by 4 cm by 5 cm.	**C.** 60 cm³	$V = lwh = 3 \cdot 4 \cdot 5$ 60 cm³	The lollipops are all solid sugar, so nothing could be wrapped inside.
5	Find the surface area of a cube with edge length 5 cm.	**A.** 150 cm²	6 faces, each $5 \cdot 5 = 25$ $6 \cdot 25$ 150 cm²	Gus Barton was loading trucks at the Loading Dock. The dock camera shows him.
6	Find the volume of a cylinder with radius 2 cm and height 5 cm. Leave your answer in terms of π.	**D.** 20π cm³	$V = \pi r^2 h = \pi(2^2)(5)$ 20π cm³	The chocolate river filters showed nothing floating. No recipe came through.
7	Find the volume of a cone with radius 3 cm and height 4 cm. Leave your answer in terms of π.	**B.** 12π cm³	$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(9)(4)$ 12π cm³	Lolly Park was taste-testing mint chips in the Test Kitchen. The sign-in sheet shows it.
8	Find the volume of a sphere with radius 3 cm. Leave your answer in terms of π.	**C.** 36π cm³	$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(27)$ 36π cm³	The candy barrels were rolled and checked. They’re full of jelly beans.
9	Find the volume of a square pyramid with a 6 cm by 6 cm base and height 4 cm.	**C.** 48 cm³	$V = \frac{1}{3}Bh = \frac{1}{3}(36)(4)$ 48 cm³	Inspector Mint was checking labels on the Packing Floor, with a clipboard full of notes.
10	Find the surface area of a rectangular prism that is 3 cm by 4 cm by 5 cm.	**A.** 94 cm²	$SA = 2(3{\cdot}4 + 3{\cdot}5 + 4{\cdot}5)$ $2(12 + 15 + 20) = 2(47)$ 94 cm²	The sample bags were counted. All held only samples.

Wrong-Answer Map

The Candy Factory Caper · What a wrong choice looks like, and the false clue it leads to

Card	Wrong choice	Likely error	False clue
1	A. 15 cm²	halved twice	The gumball machine was emptied onto a tray. Just red, blue, and green gumballs.
1	B. 16 cm²	added base and height	Mr. Sprinkle was greeting guests in the Gumdrop Garden. The guest book shows the time.
1	C. 60 cm²	forgot the half	Dr. Fizz was taking a sample by the Chocolate River. A tour guide waved to her.
2	B. 24 cm²	used one base only	Ms. Caramel was picking gumdrops in the Gumdrop Garden. Her basket was full of them.
2	C. 40 cm²	used larger base	The coin slot on the gumball machine was jammed. Nothing could be dropped in.
2	D. 64 cm²	forgot the half	Dr. Fizz was testing taffy stretch in the Taffy Room. Ms. Caramel timed her.
3	A. 3π cm²	used πr	Gus Barton was parking his forklift near the Gumdrop Garden. The yard sensor logged it.
3	C. 18π cm²	used 2πr squared factor wrong	Every gumball machine was opened with the owner's key. They held only gumballs.
3	D. 6π cm²	used circumference	Dr. Fizz was labeling boxes on the Packing Floor. The scanner log has her badge.
4	A. 12 cm³	added dimensions	Dr. Fizz was signing a delivery slip at the Loading Dock. The driver kept the copy.
4	B. 94 cm³	used surface area	Inspector Mint was inspecting the hedges in the Gumdrop Garden. His clipboard has the time.
4	D. 20 cm³	one face area	The gumball machine in the lobby was weighed. It matched the gumball count exactly.
5	B. 25 cm²	one face only	Lolly Park was sampling sour gumdrops in the Gumdrop Garden. The sample cup was empty.
5	C. 30 cm²	multiplied edge by 6	The gumball machine has a glass front and nothing was seen inside. No recipe in there.
5	D. 125 cm²	used volume	Dr. Fizz was mixing new flavors in the Test Kitchen. Lolly Park tasted each batch.
6	A. 50π cm³	swapped r and h	Gus Barton was lifting pallets and passed by the Gumdrop Garden. A camera caught his forklift.
6	B. 80π cm³	used diameter as radius	The gumball machines were all wiped and checked. Only gumballs inside.
6	C. 10π cm³	used πrh	Dr. Fizz was stirring a pot at the Chocolate River with a long paddle. Her sleeves are still sticky.
7	A. 16π cm³	swapped r and h	The gumball machine was shaken and tipped over. Not one thing but gumballs rolled out.
7	C. 4π cm³	used πrh over 3	Gus Barton was hauling a crate through the Gumdrop Garden when the bell rang. The sensor log shows it.
7	D. 36π cm³	forgot one third	Dr. Fizz was pulling sheets of taffy in the Taffy Room. The room's timer clip shows the minute.
8	A. 27π cm³	used πr³	Mr. Sprinkle was watering the sugar-plants in the Gumdrop Garden. Visitors took pictures.

8	B. 108π cm³	used $4\pi r^3$	Dr. Fizz was stacking cartons on the Packing Floor. A supervisor counted her boxes.
8	D. 12π cm³	used $\frac{4}{3}\ \pi r^2$	The gumball machine lock was sealed with tape. The tape was still intact.
9	A. 144 cm³	forgot one third	Dr. Fizz was waving off the delivery truck at the Loading Dock. The gate log has the time.
9	B. 12 cm³	forgot to use height	Ms. Caramel was dipping strawberries and then walked through the Gumdrop Garden. Her apron was spotless.
9	D. 72 cm³	used one half	A flashlight was shone into the gumball machine. It was all gumballs.
10	B. 60 cm²	used volume	Inspector Mint was measuring the paths of the Gumdrop Garden with a wheel. The wheel counter agrees.
10	C. 70 cm²	omitted a pair of faces	The gumball machine in the break room was opened. Only stale gumballs, nothing else.
10	D. 47 cm²	forgot the 2	Dr. Fizz was timing a cookie bake in the Test Kitchen. The oven display shows the time.